Dr. Anika Sharma

Scientist E — Atmospheric Instrumentation & Applied AI

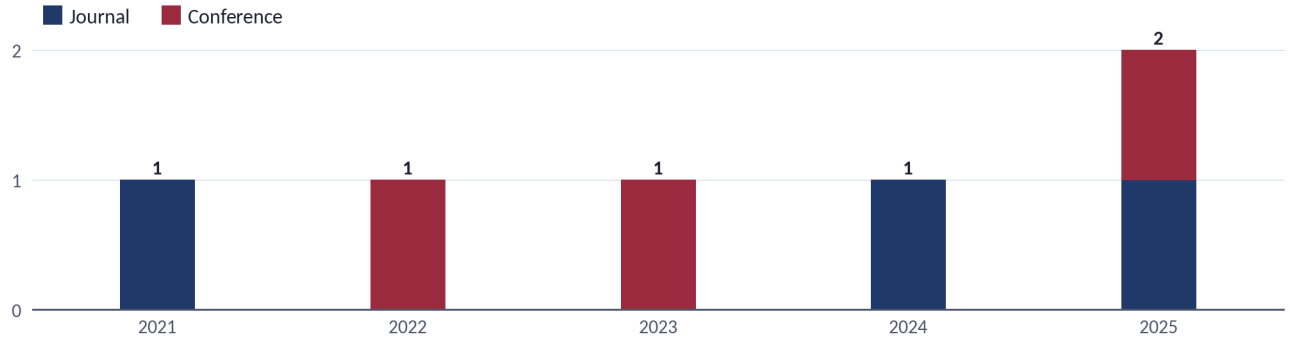
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Research Profile

Electronics and avionics engineer with twelve years of airborne-systems integration and qualification testing, followed by ten years leading scientific instrumentation and edge-AI development in government service. Experienced in MIL-STD-1553 test rigs, UAV payload electronics, sensor telemetry, embedded GPU deployment and technical procurement.

Research at a Glance

3 Journal articles	**3** Conference papers	**1** Manuscript under review	**1** Research grant
1 Talk / presentation	**142** Citations *Google Scholar (fictional example), Oct 2026*	**6** h-index *Google Scholar (fictional example), Oct 2026*	**4** i10-index *Google Scholar (fictional example), Oct 2026*



Published and accepted outputs by year (from your publication records).

Peer-reviewed Publications

[1] **A. Sharma** and R. Iyer, "Knowledge-distilled lightweight networks for drone image super-resolution," accepted for publication in Example Transactions on Aerial Imaging.

[2] **A. Sharma**, K. Rao and R. Iyer, "Benchmarking embedded GPUs for real-time image enhancement," Example Symposium on Applied AI, pp. 88-93, Jan 2025.

[3] **A. Sharma** and R. Iyer, "Channel-pruned residual networks for real-time aerial image enhancement on embedded GPUs," Journal of Embedded Vision Systems, vol. 12, no. 3, pp. 45-58, Mar 2024. doi: 10.5555/jevs.2024.0123

[4] **A. Sharma**, P. Rao and S. Gill, "An IoT telemetry framework for upper-air ground stations," Proceedings of the Example Conference on Instrumentation (ECI 2023), pp. 112-117, Sep 2023.

[5] **A. Sharma** and S. Verma, "Edge telemetry dashboards for upper-air ground stations," Proceedings of the Example Conference on Instrumentation, Paper 17, Feb 2022.

[6] **A. Sharma**, P. Menon and R. Iyer, "A low-cost GNSS receiver chain for radiosonde tracking," Example Journal of Atmospheric Instrumentation, vol. 8, no. 2, pp. 101-112, Jun 2021. doi: 10.5555/ejai.2021.0042

Manuscripts Under Review / In Preparation

[1] **A. Sharma**, M. Khan and R. Iyer, "Detection-aware quality assessment of super-resolved drone images," under review at Example Transactions on Geoscience Imaging. *[Under review]*

[2] **A. Sharma**, "Simulation study of balloon-released gliding sondes," in preparation. *[In preparation]*

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Research Grants & Funding

Seed grant: edge-AI enhancement of aerial imagery *(Completed)* Apr 2023 – Mar 2025

Principal investigator, Example Science Foundation (fictional)

Two-year seed project on lightweight super-resolution networks for embedded GPUs.

Reference: ESF/SEED/2023/017

Talks & Presentations

Invited talk: Edge AI for atmospheric observation 2024

Example Symposium on Applied AI

Appointments

Scientist E Jul 2016 – Present

Upper Air Instrumentation Division, Kestrel Atmospheric Instrumentation Laboratory, New Delhi

Responsible for upper-air observation instruments, the division workshop and the laboratory's IoT and AI/ML activities.

- Lead a four-member AI/ML group developing lightweight super-resolution and detection models for UAV imagery, deployed on NVIDIA Jetson devices with TensorRT.
- Led acceptance testing and commissioning of radiosonde ground-receiving systems at three observing stations, writing the test protocol and the defect-tracking procedure.
- Designed and deployed an IoT monitoring network for laboratory infrastructure (ESP32 sensor nodes, MQTT, Node-RED, InfluxDB and Grafana dashboards) covering temperature, humidity and power quality.
- Prepared technical specifications and served on bid-evaluation committees for instrument procurement through the Government e-Marketplace in line with the General Financial Rules.
- Authored standard operating procedures for pre-flight checks and calibration of upper-air instruments used across the division.

Additional charge: **Workshop In-charge**, Upper Air Division Workshop Apr 2019 – Present

Additional charge of the division workshop supporting instrument repair, fabrication and calibration.

- Supervise technicians and plan preventive maintenance for workshop test equipment.
- Introduced a job-card and spares-inventory system for instrument repairs and calibration work.

Nodal officer: **Nodal Officer, IoT and AI/ML Initiatives** Jan 2021 – Present

Senior Design Engineer Jun 2004 – Jun 2016

Avionics Integration Group, Halcyon Avionics Design Bureau, Bengaluru

Integration and qualification testing of airborne electronics and UAV payloads.

- Developed payload interface electronics for a fixed-wing UAV electro-optical sensor and supported its ground and flight trials.
- Integrated mission-computer and display units on MIL-STD-1553 avionics buses and built the automated test rig used for bench validation.
- Planned environmental qualification (MIL-STD-810 vibration, temperature and humidity) for line-replaceable units and documented the test results.
- Mentored graduate engineer trainees on avionics bus protocols and test documentation.

Graduate Engineer Aug 2002 – May 2004

Test Systems Group, Meridian Electronics Limited, Bengaluru

Built automated test equipment for avionics line-replaceable units.

- Developed LabVIEW test sequences for power-supply and interface boards of avionics units.

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Research Projects

Edge super-resolution pipeline for UAV imagery *(Ongoing)* — **Sep 2021 – Present**

Principal investigator, Kestrel Atmospheric Instrumentation Laboratory

Research prototype that enhances low-resolution UAV video in real time on embedded GPUs using pruned and distilled deep networks.

- Implemented structured channel pruning and knowledge distillation to shrink the super-resolution network for embedded deployment.
- Converted models through ONNX to INT8 TensorRT engines, achieving 3.1× faster inference on Jetson Xavier NX.

Technologies: NVIDIA Jetson Xavier NX, NVIDIA Jetson Orin Nano, PyTorch, TensorRT, ONNX, Docker, Structured pruning, Knowledge distillation, INT8 quantisation

Radiosonde ground-station acceptance test protocol — **Jan 2018 – Mar 2019**

Coordinator

Challenge: Station acceptance testing varied between sites.

Outcome: A common acceptance protocol and checklist adopted for new ground-receiving systems.

IoT monitoring of laboratory infrastructure — **Feb 2020 – Jun 2022**

Project lead

Challenge: Instrument rooms lacked continuous environmental and power-quality monitoring.

Approach: Deployed ESP32 sensor nodes publishing over MQTT to a Node-RED flow, InfluxDB time-series storage and Grafana dashboards with threshold alerts.

Outcome: Continuous monitoring with automated alerts replaced manual twice-daily readings.

Technologies: ESP32, Environmental sensors, MQTT, Node-RED, InfluxDB, Grafana

Education

Ph.D., Electrical and Electronics Engineering — 2021 – ongoing

Northfield Institute of Science and Technology

Thesis: Lightweight deep networks for real-time enhancement of aerial imagery

M.Tech., Avionics — 2010

Coastal Institute of Aeronautical Engineering, Chennai

Grade: CGPA 8.6/10

Thesis: Fault-tolerant data acquisition for airborne test instrumentation

B.E., Electronics and Communication Engineering — 2004

Northfield Institute of Science and Technology, Northfield University, Pune

Grade: First Class with Distinction

Professional Service

Reviewer, Example Journal of Atmospheric Instrumentation — 2022 – Present

Example Journal of Atmospheric Instrumentation

Co-supervisor, two M.Tech. dissertations in embedded vision — 2024 – 2025

Example Institute of Technology (fictional)

Awards & Honours

Best Paper Award — 2023

Example Conference on Instrumentation (ECI 2023)

Professional Memberships

Member, Institution of Example Engineers — 2010 – Present

Institution of Example Engineers

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Technical Skills

Avionics
MIL-STD-1553 / ARINC 429

Avionics integration

Scientific instrumentation
Environmental qualification testing

Sensor calibration

Radiosonde and upper-air systems

Embedded systems
Embedded C

ESP32 / STM32 microcontrollers

Computer vision
Image super-resolution

Object detection

Drones & UAV systems
UAV payload integration

Edge AI
NVIDIA Jetson deployment

AI / machine learning
Deep learning

PyTorch

Procurement & administration
Technical procurement (GeM, GFR)

IoT
MQTT and Node-RED

InfluxDB and Grafana

Model optimisation
TensorRT and ONNX

Pruning and knowledge distillation

Software & tools
Python

Docker and Linux

Technical leadership
Team leadership

GPU computing
CUDA / GPU computing

Bars show my own assessment of proficiency (five steps; five = expert).

Referees

Dr. R. Iyer (fictional)
Professor of Electronics, Example Institute of Technology
Research collaborator and co-author
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